

1.



SIMATS
ENGINEERING

SIMATS Online Programming Lab

C++ PROGRAMMING

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QUESTIONS 9 / 20 completed

Shape Hierarchy ▾

SHAPE HIERARCHY

Shape Hierarchy:
Create a hierarchy of shapes (e.g., square, circle, triangle) using a base class and derive specific shapes with methods to calculate area and display shape properties.
Sample Input:
Enter the dimensions of the square:
Side: 5

PROGRAM EDITOR

C++ ▾

Run Save

```
1 #include <iostream>
2 #include <vector>
3 #include <string>
4
5 using namespace std;
6
7 class Item {
8 public:
9     string name;
10    double price;
11    int quantity;
12
13    Item(string n, double p, int q) : name(n), price(p),
14 };
15
16 class ShoppingCart {
17 private:
18     vector<Item> cart;
19
20 public:
```

INPUTHeadphones
50
2

2.



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QUESTIONS 16 / 20 completed

Online Shopping C

ONLINE SHOPPING CART
Online Shopping Cart:
Design a class for an online shopping cart that includes products, their prices, and quantities. Implement functions to add items, remove items, and calculate the total cost.
Sample Input:
Adding items to the

PROGRAM EDITOR C++

Run
Save

```

1 #include <iostream>
2 #include <string>
3 #include <vector>
4
5 using namespace std;
6
7 class Product {
8 public:
9     string name;
10    double price;
11    int quantity;
12
13    Product(string n, double p, int q) : name(n), price(p), quantity(q) {}
14 };
15
16 class ShoppingCart {
17 private:
18     vector<Product> cart;
19
20 public:
21

```

INPUT

```

Laptop
1000
2
Laptop
1

```

simatslab.saveetha.com/practice_app/dashboard_student

3.



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QUESTIONS 16 / 20 completed

School Manageme

SCHOOL MANAGEMENT SYSTEM
School Management System: Develop a class hierarchy to manage a school system. Create classes for students, teachers, courses, and implement functionalities to enroll students, assign teachers, etc.
Sample Input:
Adding a new student.
Student Name: John

PROGRAM EDITOR C++

Run
Save

```

1 #include <iostream>
2 #include <string>
3 #include <vector>
4
5 using namespace std;
6
7 class Student {
8 public:
9     string name;
10    string id;
11    Student(string n, string i) : name(n), id(i) {}
12 };
13
14 class Teacher {
15 public:
16     string name;
17     string id;
18     Teacher(string n, string i) : name(n), id(i) {}
19 };
20
21

```

INPUT

```

John Doe
12345
Jane Smith
T1001
12345
CSCI101
T1001
CSCI101

```

4.



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QUESTIONS 16 / 20 completed

Banking System with Multiple Accounts

BANKING SYSTEM WITH MULTIPLE ACCOUNTS

Banking System with Multiple Accounts: Extend the bank account class to handle multiple accounts, allowing functions for transfer of funds between accounts and viewing all account details.

Sample Input:
Creating multiple

PROGRAM EDITOR
C++
Run Save

```

1 #include <iostream>
2 #include <string>
3 #include <vector>
4
5 using namespace std;
6
7 class BankAccount {
8 public:
9     string accountNumber;
10    string accountHolder;
11    double balance;
12
13    BankAccount(string accNum, string holder, double ini
14                : accountNumber(accNum), accountHolder(holder),
15 };
16
17 class Bank {
18 private:
19     vector<BankAccount> accounts;
20
21

```

INPUT

```

1234567890
John Doe
5000
0987654321
Jane Smith
3000
1234567890
0987654321
2000

```

5.



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QUESTIONS 16 / 20 completed

Movie Database

MOVIE DATABASE

Movie Database: Design a class representing a movie database with attributes like title, director, genre, and release year. Include methods to add, delete, and display movie details.

Sample Input:
Adding a new movie to

PROGRAM EDITOR
C++
Run Save

```

1 #include <iostream>
2 #include <string>
3 #include <vector>
4
5 using namespace std;
6
7 class Movie {
8 public:
9     string title;
10    string director;
11    string genre;
12    int releaseYear;
13
14    Movie(string t, string d, string g, int y)
15        : title(t), director(d), genre(g), releaseYear(y
16 };
17
18 class MovieDatabase {
19 private:
20     vector<Movie> database;
21

```

INPUT

```

Inception
Christopher Nolan
Science Fiction
2010
Inception
Inception

```